

Module 4 — Stretch: Visualization Critique

Honors Track. This stretch assignment is not required for program completion but counts toward Honors distinction. Only work on this if you have completed all core assignments for this module and are On Track or Advanced. If you are behind on core work, focus there first.

The Challenge

Find 3 published data visualizations — from news sites, government reports, academic papers, or data journalism — and analyze them critically using the chart type selection framework and Gestalt principles from the Module 4 reading.

This is a professional communication skill. Data scientists don't just create visualizations — they evaluate them. Understanding what makes a chart effective (or misleading) sharpens your own design judgment.

What You'll Learn

- Critical evaluation of chart type choices
- Identifying Gestalt principles in practice (proximity, similarity, enclosure, connection)
- Recognizing misleading visual encodings (truncated axes, inappropriate scales, cherry-picked ranges)
- Articulating design improvements with specific, actionable suggestions

Tasks

1. Find 3 visualizations

Select 3 data visualizations from different sources. At least one should be a chart you think is **effective** and at least one should be a chart you think is **problematic**. Good sources:

- News sites: Al Jazeera Data, Reuters Graphics, The Economist, FiveThirtyEight
- Government: Jordan Department of Statistics, World Bank Open Data, UN Data
- Academic: papers with figures from Google Scholar
- Data journalism: The Pudding, Our World in Data, Flowing Data

Save or screenshot each visualization.

2. Analyze each visualization

For each of the 3 visualizations, write a critique covering:

1. **Source and context:** Where did you find it? What data does it show? Who is the intended audience?
2. **Chart type assessment:** Is the chart type appropriate for the data and the question it's trying to answer? What alternatives could work? Reference the chart type selection framework (comparison → bar, trend → line, distribution → histogram, relationship → scatter, etc.)
3. **Gestalt principles:** Which Gestalt principles are at work? Does the visual grouping help or hinder comprehension?
4. **Color and accessibility:** Is the color palette effective? Would it work for colorblind viewers?
5. **Labels and annotation:** Does the chart have a finding-based title (not just a data description)? Are axes labeled with units? Are key points annotated?
6. **Overall verdict:** Effective, partially effective, or misleading? One sentence on why.
7. **Your improvement:** If you could redesign this chart, what would you change? Be specific.

3. Write your critique document

Create a Markdown file `VISUALIZATION_CRITIQUE.md` with your analysis of all 3 visualizations. Include the image or a link to the source for each.

4. Submit

Submit your `VISUALIZATION_CRITIQUE.md` by pasting the file content or a link into TalentLMS → Module 4 → Tuesday Stretch.

How This Assignment Is Graded

Your TA scores this assignment on 3 dimensions: Source Selection, Critique Depth, and Actionable Improvements. What matters most is the specificity and depth of your analysis — not the length.

AI screening: Your TA screens submissions for generic, AI-generated content. Submissions flagged as AI-generated receive a **0 with resubmission required**. The critique is hard to fake well because it requires you to describe specific visual elements in specific charts. Generic observations like “the chart effectively uses color to convey meaning” are an immediate red flag — name the color, name the element, explain why it works or doesn't.

Tips

- Don't just pick obviously bad charts. The most interesting critiques are of charts that are *almost* good but have a specific design choice that weakens them.
- Be specific in your improvements. “Make it better” is not a critique. “Replace the pie chart with a horizontal bar chart sorted by value, because the audience needs to compare 8 categories and pie slices are hard to compare beyond 3–4 segments” is a critique.
- Reference the reading's terminology: Gestalt principles, colorblind-safe palettes, finding-based titles, chart type selection framework.